

Research Assignment

Data Normalization Brief

Data normalization is an important preprocessing step in machine learning that helps to bring all features to a similar scale so that no single feature dominates the model.

Z-score Standardization

Standard Scaler converts data using mean and standard deviation:

$$Z = (x - \mu) / \sigma$$

It shifts the data so that:

- Mean becomes 0
- Standard deviation becomes 1

This method is based on the distribution of data and keeps the shape of the data unchanged, but it does not restrict values to a fixed range.

Min-Max Scaler

Min-Max Scaler converts data into a fixed range, usually between 0 and 1:

$$x' = \{x - x_{\min}\} / \{x_{\max} - x_{\min}\}$$

It compresses all values into a limited range and depends only on minimum and maximum values. It is more sensitive to outliers because extreme values affect the scaling.

Main Difference

Standard Scaler works based on **mean and variance**, while Min-Max Scaler works based on **minimum and maximum values**. Standard Scaler standardizes the distribution, whereas Min-Max only rescales the range of data.

Why wrong scaling affects SVM

Support Vector Machines (SVM) depend on distance between data points to find the best separating hyperplane. If features are not properly scaled:

- Features with large values dominate the distance calculation
- The decision boundary becomes biased
- The model performance decreases

For kernel-based SVM (like RBF), incorrect scaling can completely distort the similarity between points, leading to poor accuracy and wrong predictions.

Conclusion

Standard Scaler is better when the data distribution matters, especially for models like SVM, while Min-Max Scaler is useful when we need all features within a fixed range. Choosing the wrong scaling method can lead to incorrect model behavior and reduced performance.